1 2 9 0 FACULDADE DE CIÊNCIAS E TECNOLOGIA UNIVERSIDADE D COIMBRA

deer.uc DEPARTAMENTO DE ENGENHARIA ELETROTÉCNICA E DE COMPUTADORES

Master's Degree in Electrical and Computer Engineering

DIGITAL SYSTEM PROJECT

PROJECT TITLE

4-bit ALU Calculator - VHDL Project Report

Mark John Ensa – 2024251008

Hina Iqbal - 2023182290

Introduction/Project Overview

This project implements a **4-bit ALU (Arithmetic Logic Unit) calculator** using Verilog on the Altera DE2 development board featuring the Cyclone II FPGA (EP2C35F672C6). and graphical block-based design with **Intel Quartus II**. The system takes input via a **PS/2 keyboard**, processes the operations through custom-designed digital logic, and outputs the result on a **VGA monitor** and **7-segment display**.

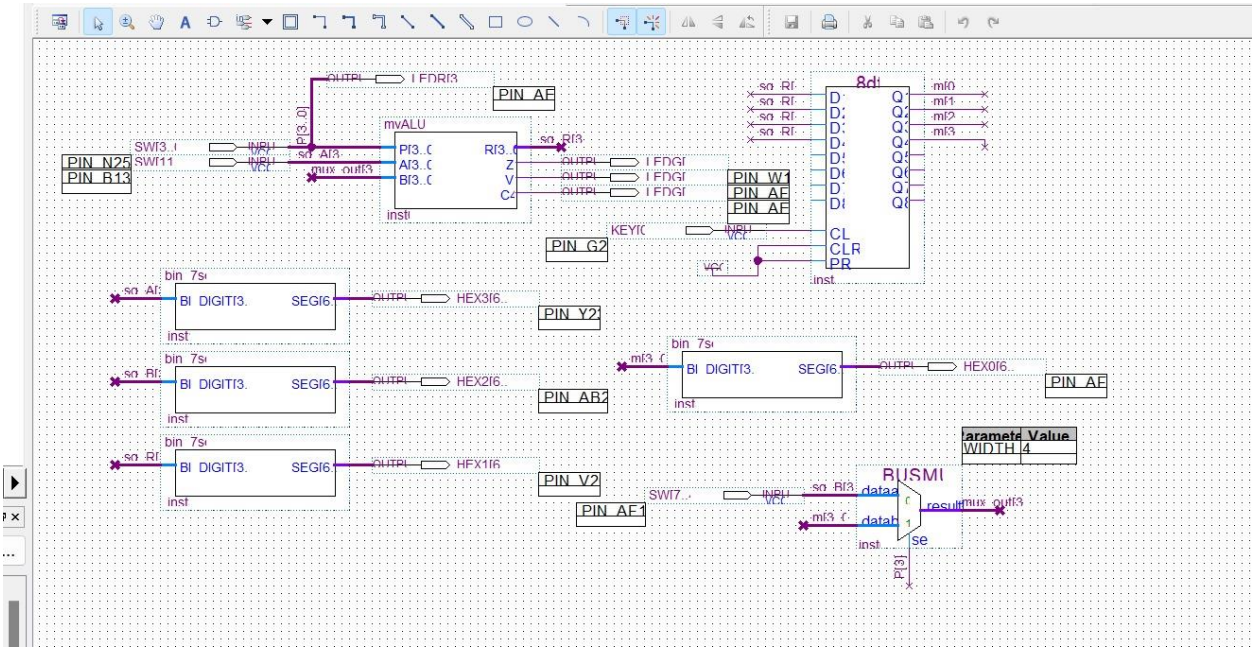
The 4-bit ALU (Arithmetic Logic Unit) Calculator is a digital system designed to perform basic arithmetic and logic operations on two 4-bit binary inputs. Implemented using Verilog HDL and deployed on the Altera DE2 FPGA development board, this project serves as a foundational exercise in digital design, hardware description languages, and FPGA prototyping.

The ALU accepts two 4-bit operands and supports four fundamental operations: addition, subtraction, bitwise AND, and bitwise OR. Users interact with the system through hardware switches on the DE2 board to provide input operands and select the desired operation. The output result is displayed using the board's LED indicators, providing real-time, visible feedback of the ALU's functionality.

This project illustrates core principles of digital systems design, such as combinational logic, modular hardware development, and binary arithmetic. It also demonstrates the practical workflow of FPGA-based development, including design entry in Verilog, synthesis and implementation using Intel Quartus, and hardware testing on an actual FPGA platform.

By combining theoretical understanding with hands-on hardware implementation, the 4-bit ALU Calculator provides valuable learning experience in the field of digital electronics and embedded systems.

The design modularizes core arithmetic functions such as binary addition and logical operations. It is ideal for deployment on FPGA development boards such as the **Altera DE2 board**.



Project File Structure

The project folder contains the following main files:

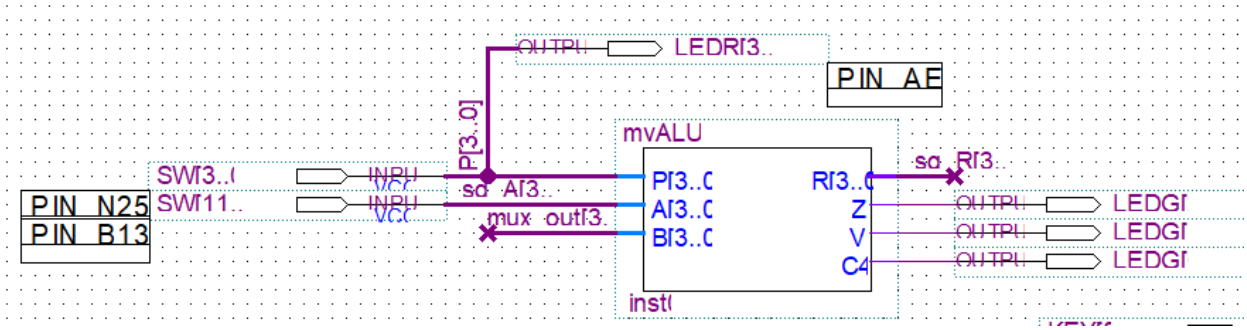
- Calculator.qpf – Quartus Project File.
- Calculator.qsf, Calculator.qws – Project configuration and workspace files.
- .bdf files – Block Diagram Files for graphical HDL design.
- .v files – Verilog source code modules.
- .bsf files – Block Symbol Files used in schematic designs.
- /db/ folder – Internal Quartus database and compilation files (not user-editable).

Block Diagram Structure

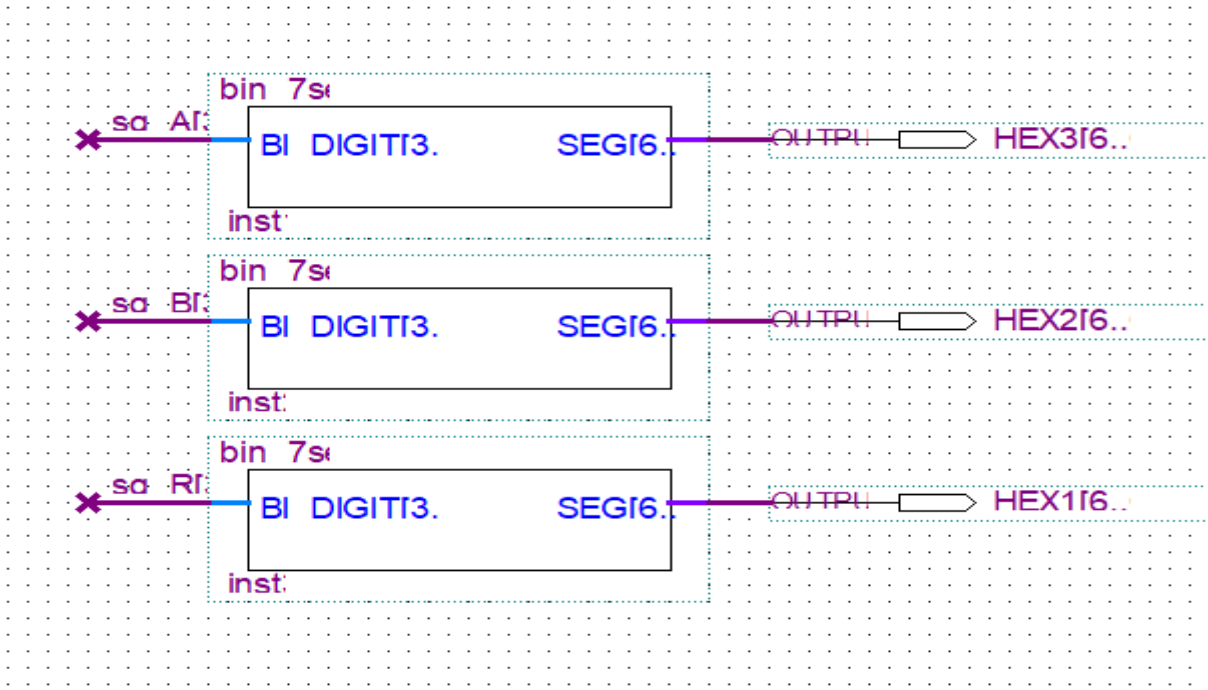
The system is composed of the following core modules:
- PS/2 Keyboard Interface: Captures hexadecimal and control inputs.
- ALU Core (Verilog): Executes arithmetic and logic operations on 4-bit data.
- VGA Sync Generator: Generates 640x480 VGA timing.
- Display Renderer: Displays operands, operations, and results on-screen.
- Top-Level Module: Instantiates and connects all components.

Key Modules and Descriptions

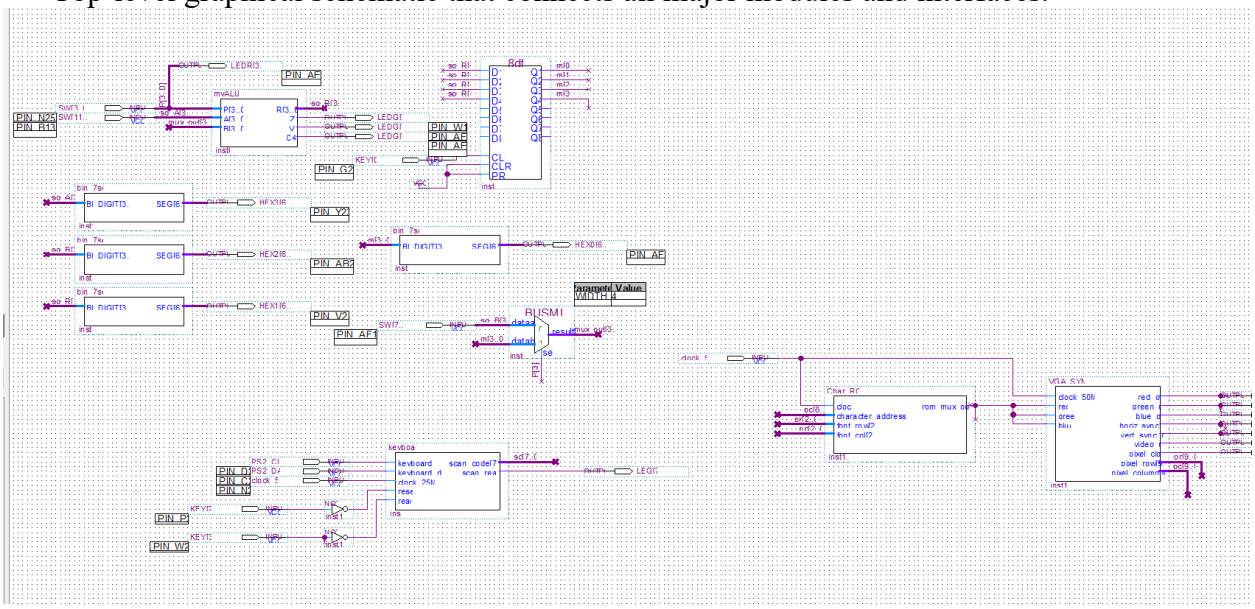
- **myALU4.v/myALU4.bsf**
  Implements the main 4-bit ALU. Supports operations such as addition, subtraction, AND, OR, etc., depending on control inputs.



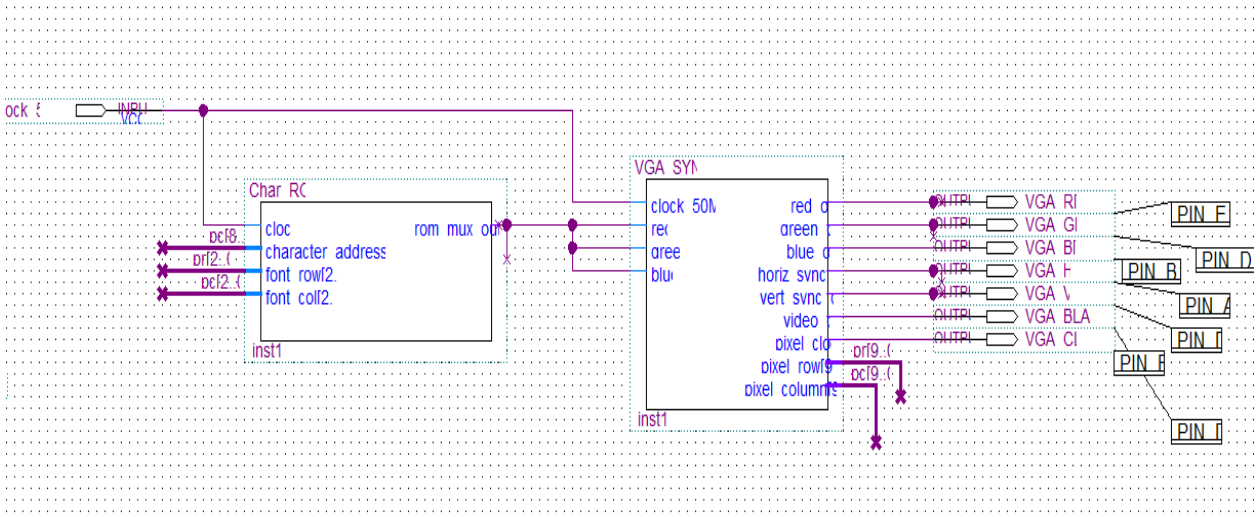
- **myadder4.v/myadder4.bsf**
  Performs 4-bit binary addition using four instances of the full adder module (myfulladd).
- **myfulladd.v/myfulladd.bsf**
  Basic 1-bit full adder logic with inputs A, B, and Cin (carry in), producing sum and carry out.
- **bin_7seg.v/bin_7seg.bsf**
  Converts 4-bit binary numbers into control signals for a 7-segment display.



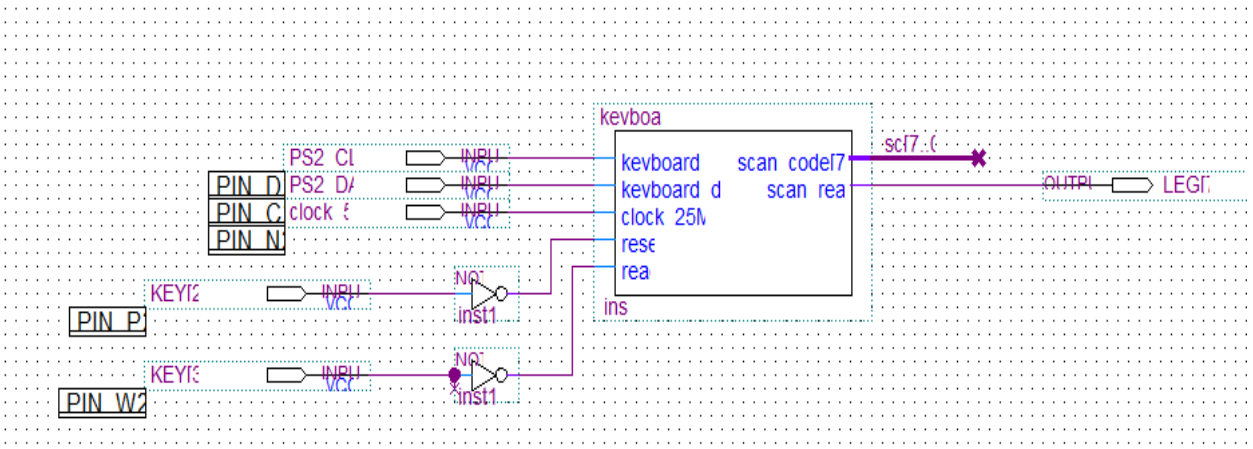
- **Calculator.bdf**
  Top-level graphical schematic that connects all major modules and interfaces.



- **Char_ROM/VGA_SYNC.bdf**
  Handles character rendering and synchronization signals for displaying information on a VGA monitor. Provides horizontal and vertical synchronization signals for VGA display at 640x480. Uses a 25 MHz clock derived from the DE2's 50 MHz system clock. Pixel coordinates (x, y) are generated.



- **Keyboard/dec_7seg_b.bdf**
  Integrates PS/2 keyboard input processing and 7-segment decoding logic. The PS/2 module captures keyboard scan codes and converts them into usable inputs. It recognizes numeric keys (0-9), operation keys (+, -, *, /), and control signals for result confirmation.



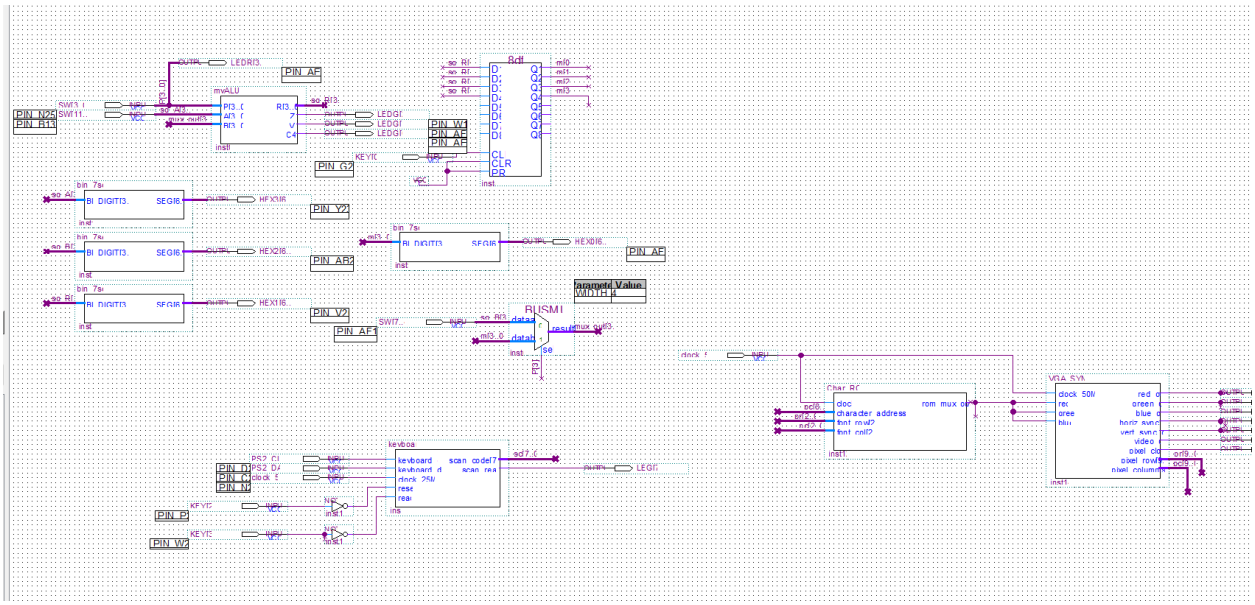
Usage Instructions

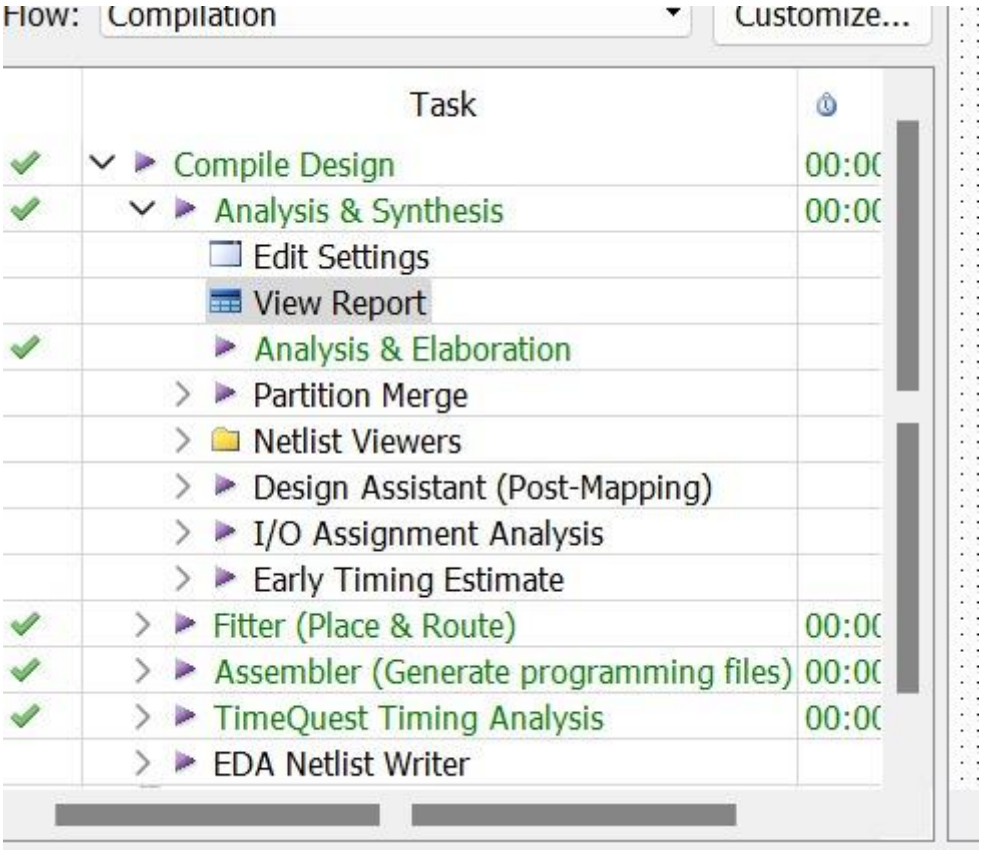
To use this project on an FPGA board:

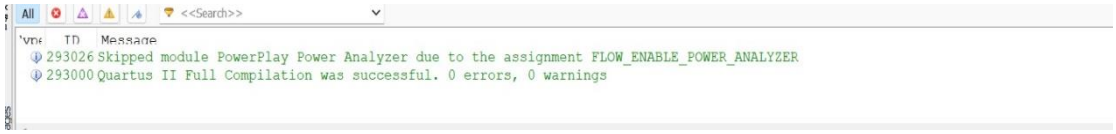
1. **Open Quartus II** and load the Calculator.qpf file.



2. Compile the project using the **"Start Compilation"** option.







3. Connect the FPGA board (e.g., DE2-115) via USB.
4. Use the **Programmer** tool in Quartus to load the compiled .sof file to the FPGA.
5. Attach a **PS/2 keyboard** and **VGA monitor** to the board.
6. Press keys to input data and operations. Observe the results on both:

- The **VGA screen** (via Char_ROM and VGA_SYNC).
- The **7-segment display** (via bin_7seg).

Top-Level Integration

The top-level module integrates PS/2, ALU, VGA Sync, and Display logic. It connects DE2 board input/output pins:
- PS/2 Clock and Data
- VGA Red, Green, Blue, HSync, VSync
- Onboard clock source and reset button

User Interaction

The user provides 4-bit inputs through the keyboard. Sequence:
- Enter operand A digits
- Enter operand B digits
- Select operation (+, -, &, |, etc.)
- Result is displayed immediately Example: '2', '3', '+', '1', '2' calculates 0x23 + 0x12

Summary

This Verilog-based ALU calculator demonstrates real-time hardware interaction on the DE2 FPGA board. It integrates digital design fundamentals, keyboard interfacing, video signal generation, and modular Verilog HDL design. It functions entirely on hardware without the use of simulation tools. This 4-bit ALU calculator project showcases the integration of custom logic modules with graphical HDL design. It demonstrates concepts such as:

- Modular Verilog design
- Hardware-level arithmetic processing
- Input handling via PS/2 interface
- Output rendering through VGA and 7-segment displays

It provides a solid platform for learning and experimenting with digital design and FPGA-based systems.